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DOCUMENT CLASS: TECHNICAL PROOF BRIEF · FOR INSTITUTIONAL REVIEW

ENTERPRISE SATELLITE CONNECTIVITY · GOVERNANCE CONTROL PLANE

INTELLILINK GATEWAY™

Technical Proof Brief
Enterprise Satellite Connectivity Governance Control Plane

DOCUMENT	VERSION	DATE	STATUS
Technical Proof Brief	2.0	07 February 2026	Technical Validation Summary

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DOCUMENT CONTROL

FIELD	DETAILS
Document Title	Intellilink Gateway™ — Technical Proof Brief

FIELD	DETAILS
Document Type	Technical Validation Summary
Version	2.0
Date	07 February 2026
Status	Technical Validation Summary
Prepared by	Intellilink Media LLC™ (USA)
Distribution	Institutional · Regulatory · Technical Stakeholders
Classification	Public — Institutional Release (Anonymized)
Network Identifiers	No ISP names, locations, or sensitive network identifiers disclosed

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SECTION 01

Executive Technical Overview

The rapid adoption of satellite Internet services, particularly low-Earth-orbit (LEO) systems, has significantly improved connectivity resilience for enterprises across Africa. However, many enterprise deployments operate outside traditional Internet Service Provider (ISP) delivery models, reducing visibility, accountability, and alignment with national Internet governance frameworks.

Intellilink Gateway™ introduces a control-plane architecture designed to restore traditional ISP governance characteristics — including local IP anchoring, accountable upstream presence, and auditability — while preserving the performance and resilience benefits of satellite connectivity.

This document summarizes both laboratory and field validation of the Intellilink Gateway™ control plane under real operating conditions.

SECTION 02

Architectural Overview

The Intellilink Gateway™ architecture separates access connectivity from governance enforcement.

ACCESS LAYER (UNCHANGED)

- Satellite transport underlay (e.g., Starlink)
- Enterprise retains direct satellite performance characteristics

GOVERNANCE LAYER (INTRODUCED)

- Secure gateway node adjacent to satellite CPE
- Encrypted tunnel to a domestic ISP Point of Presence (PoP)
- Policy enforcement, traffic visibility, addressing, and logging applied at the PoP

Logical Traffic Flow

```
Enterprise LAN
  → Satellite CPE (Starlink)
  → Intellilink Gateway™ (Agent Node)
  → Encrypted Tunnel
  → ISP PoP (Governance Anchor)
  → Internet
```

ARCHITECTURAL NOTE

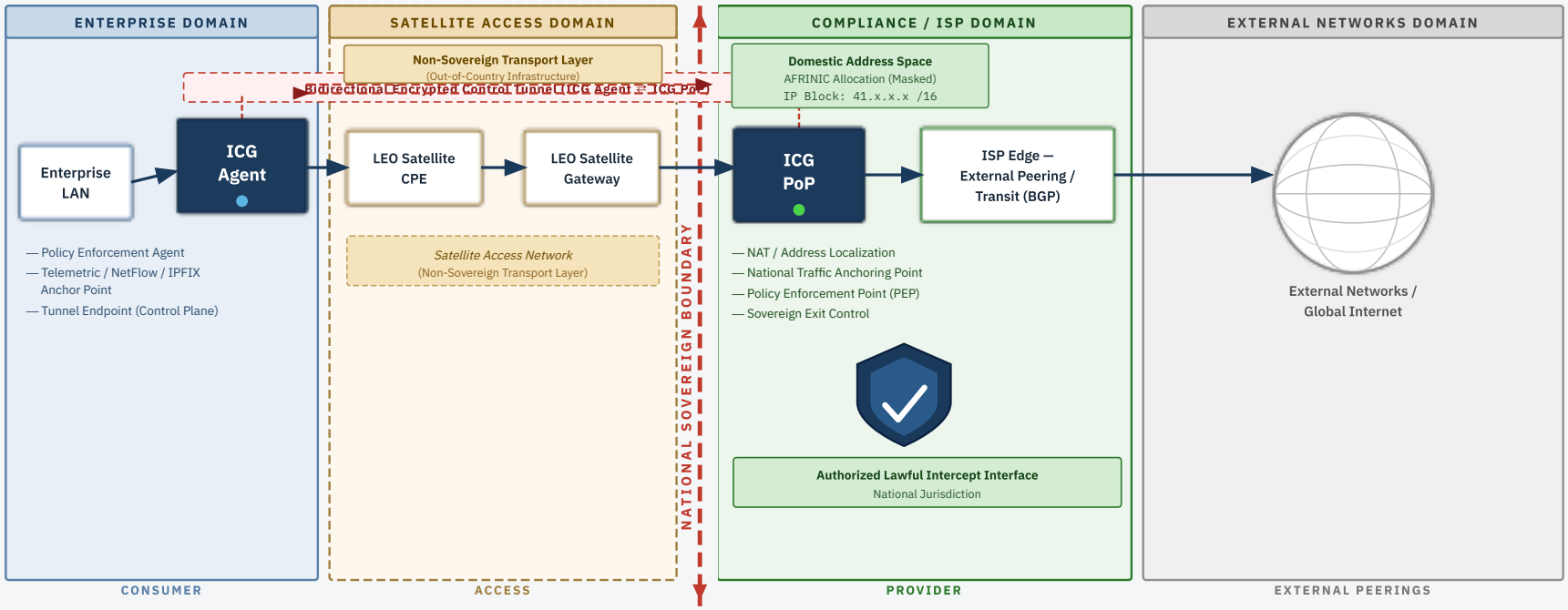
The architecture does not replace satellite connectivity. Instead, it re-anchors enterprise traffic into a conventional ISP governance domain.

USE CASE: DOMESTIC TRAFFIC ANCHORING FOR SATELLITE BROADBAND NETWORKS

Intellilink Compliance Gateway (ICG) — Reference Architecture

Architecture for compliant enterprise satellite connectivity with domestic traffic anchoring and sovereign enforcement.

Document Class: Technical Proof Brief
Revision: 3.0 | February 2026
For Regulatory Review



KEY:

- National Sovereign Boundary (jurisdictional delimiter)
- Bidirectional Encrypted Control Plane (ICG Agent ⇄ ICG PoP) — not visible in traceroute / data plane
- Data Plane — User Traffic
- Authorized Lawful Intercept (LI) Interface — National Jurisdiction
- Non-Sovereign Transport Layer (out-of-country infrastructure — encapsulated)
- Policy Enforcement Point (PEP) — ICG Agent & ICG PoP
- Domestic Address Space — AFRINIC Allocation (Masked) — 41.x.x.x/16
- ISP Edge — External Peering / Transit (BGP)

INTELLILINK COMPLIANCE GATEWAY (ICG) — REFERENCE ARCHITECTURE | TECHNICAL PROOF BRIEF
Figure 3 | Use Case: Domestic Traffic Anchoring — Satellite Broadband Networks | Rev. 3.0 | February 2026
All IP addresses masked for anonymity. No real infrastructure addresses disclosed. No vendor branding. Neutral, standards-based telecommunications terminology.

Page 1 of 1

FIGURE 1 Intellilink Gateway™ Technical Architecture — The architecture introduces a governance control plane that anchors satellite-originated enterprise traffic within a domestic ISP Point of Presence while preserving satellite access performance.

SECTION 03

Validation Scope

3.1 Laboratory Validation (Completed)

The following capabilities were validated in a controlled laboratory environment:

- Secure tunnel establishment between Gateway and ISP PoP
- Local IP address assignment under ISP-controlled address space
- Traffic forwarding consistency under tunneled operation
- NAT and routing behavior aligned with traditional ISP models
- Separation of management and traffic planes

3.2 Field Validation (Completed)

The architecture was deployed and tested in a live production environment with anonymization applied. Deployment components included:

- One domestic ISP Point of Presence (identity withheld)
- One enterprise network site
- Active satellite connectivity using Starlink
- Real enterprise traffic traversing the governance control plane

Field validation confirmed:

- Stable tunnel operation
- Correct traffic anchoring at the ISP PoP
- Transparent application behavior for enterprise users
- No material performance degradation attributable to the control layer

USE CASE: DOMESTIC TRAFFIC ANCHORING FOR SATELLITE BROADBAND NETWORKS

Intellilink Compliance Gateway (ICG)
Traffic Anchoring — Before / After Routing Comparison

Figure 2 — Traceroute-Based Routing Validation | Source: Enterprise LAN (192.168.x.x, Lusaka) → Destination: speedtest.mtn.zm (41.223.117.81)

For Regulatory Review

Document Type: Technical Proof Brief
Revision: 2.0 | Date: February 2026
Prepared For: Telecommunications Regulator
Originator: Intellilink / ICG Architecture Team

Routing Behaviour Comparison Demonstrates how ICG deployment anchors enterprise satellite traffic within national legal jurisdiction, restoring domestic policy enforcement and lawful intercept capability.

⚠ BEFORE ICG
17 hops / 242 ms
8 foreign ASN hops | No LI capability

✓ AFTER ICG
7 hops / 302 ms
0 foreign ASN hops | Full LI capability

Note on latency delta (+60 ms): *Increased latency reflects the overhead of domestic traffic anchoring, NAT processing, and compliance enforcement at the ICG PoP. This is consistent with the security and regulatory objectives of the deployment.*

LEGEND Data Plane (User Traffic) — — — Encapsulated Tunnel (Data Plane) — — — Control Plane (Encrypted — not traceroute-visible) Foreign / Non-Sovereign ASN Domestic ASN (Zambia) Encapsulated Hop (not data-plane visible) Sovereign Network Boundary

⚠ BEFORE ICG DEPLOYMENT — NON-SOVEREIGN ROUTING (NON-COMPLIANT)

DATA PLANE ORIGIN
ENTERPRISE LAN
192.168.1.79 — LUSAKA, ZAMBIA

ENTERPRISE DOMAIN
No Compliance Gateway
Traffic exits unencapsulated

SATELLITE ACCESS DOMAIN — NON-SOVEREIGN TRANSPORT
LEO Satellite CPE
Satellite Provider Network
LEO Satellite Gateway
Foreign egress — jurisdiction exit

SOVEREIGN NETWORK BOUNDARY — TRAVERSED WITHOUT CONTROL

EXTERNAL NETWORKS — INTERNATIONAL TRANSIT (FOREIGN ASNS)
Foreign ASN Transit
206.224.73.x / 154.66.247.x / 149.6.x.x
No lawful intercept · No domestic policy enforcement

↩ RE-ENTRY INTO NATIONAL JURISDICTION

DOMESTIC ISP DOMAIN (RE-ENTRY — ZAMBIA ASN)
Carrier / ISP Edge
ISP Edge

TRACEROUTE — BEFORE ICG | ENTERPRISE LAN → 41.223.117.81

1	192.168.1.1	112.817 ms	LAN Default GW
2	100.64.0.1	171.163 ms	CGNAT (Sat. CPE)
3	172.16.250.86	170.346 ms	Satellite Provider PoP
⚠ Sovereign Network Boundary crossed — foreign ASN transit begins			
4	206.224.73.68	170.073 ms	Foreign ASN
5	206.224.73.52	170.196 ms	Foreign ASN
6	102.68.115.226	170.877 ms	Foreign ASN
7	154.66.247.66	204.070 ms	Foreign transit
8	154.66.247.106	62.200 ms	Foreign transit
9	154.66.247.99	203.870 ms	Foreign transit
10	* * *		No response (filtered)
11	149.6.144.125	216.014 ms	Foreign transit
↩ Re-entry into Zambia domestic ASN			
12	41.181.105.163	215.952 ms	Carrier Edge
13	41.181.244.162	339.148 ms	ISP Core
14	41.181.244.205	201.123 ms	ISP Core
15	41.181.190.133	193.452 ms	ISP Distribution
16	41.223.119.202	238.447 ms	MTN Edge
17	41.223.117.81	242.281 ms	DESTINATION ✓

✓ AFTER ICG DEPLOYMENT — DOMESTICALLY ANCHORED ROUTING (COMPLIANT)

DATA PLANE ORIGIN
ENTERPRISE LAN
192.168.101.5 — LUSAKA, ZAMBIA

ENTERPRISE DOMAIN
ICG Agent (Compliance Gateway)
PEP TUNNEL ENDPOINT
Policy enforcement · Telemetry · Routing control

Control Plane (Encrypted) — ICG Agent ⇄ ICG PoP
Tunnel management · Policy updates · Telemetry
Not visible in traceroute output

SATELLITE ACCESS DOMAIN — NON-SOVEREIGN TRANSPORT (TUNNEL-ENCAPSULATED)
LEO Satellite CPE
Encapsulated tunnel (data plane not exposed)
Satellite Provider Network
LEO Satellite Gateway
Transit only — payload opaque to provider

✓ SOVEREIGN NETWORK BOUNDARY — TRAFFIC ANCHORED DOMESTICALLY

COMPLIANCE / ISP DOMAIN — NATIONAL POLICY ENFORCEMENT ZONE
ICG PoP — Domestic Anchor Point
NAT / ADDRESS LOCALIZATION PEP
LAWFUL INTERCEPT
National ASN egress · Zambia legal jurisdiction · AFRINIC address space
ISP Edge — National ASN (Lusaka)

TRACEROUTE — AFTER ICG | ENTERPRISE LAN → 41.223.117.81

1	192.168.101.1	2.020 ms	ICG Agent — LAN GW
→ Satellite segment (encapsulated tunnel — hops not data-plane visible)			
2	172.16.10.1	246.031 ms	ICG PoP ingress — tunnel decap
3	41.x.x.x	243.932 ms	NAT egress — Local ISP ASN (masked)
4	* * *		No response (ICMP filtered)
→ Local IXP / MTN domestic transit			
5	192.32.209.224	291.902 ms	Local IXP
6	41.223.119.202	269.292 ms	MTN Edge
7	41.223.117.81	302.121 ms	DESTINATION ✓

■ Satellite transit hops are encapsulated within the ICG tunnel and are not visible as individual traceroute hops. This is expected behaviour and does not indicate routing anomalies.

COMPLIANCE ASSESSMENT — COMPLIANT

- ✔ Zero foreign ASN hops observed in traceroute. All intermediate addresses within domestic ASN space.
- ✔ Address localization enforced at ICG PoP via NAT; public egress uses domestic ASN address space.
- ✔ Lawful intercept capability present at ICG PoP, within national legal jurisdiction.
- ✔ Policy Enforcement Point (PEP) active at both ICG Agent (enterprise)

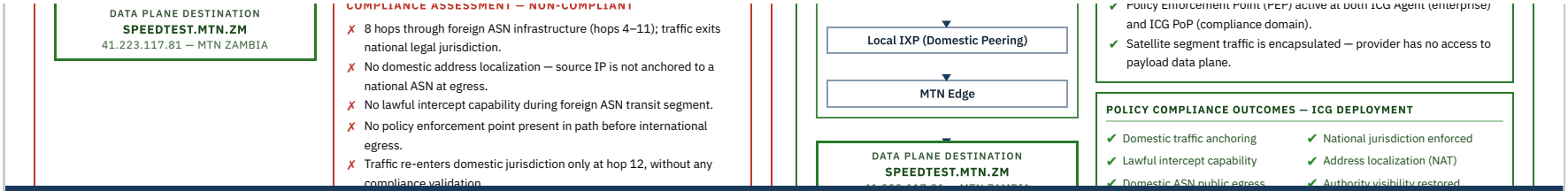


FIGURE 2 Routing Behavior Comparison Before and After Governance Anchoring — The diagram illustrates how enterprise traffic that previously exited the satellite network through external upstream paths becomes anchored within a domestic ISP governance domain after deployment of the Intellilink Gateway™ control plane.

SECTION 04

Field Environment (Anonymized)

ROLE	DESCRIPTION
Satellite Underlay	LEO satellite service (Starlink)
ISP Role	Governance anchor and PoP termination
Enterprise Role	Traffic source and destination
Gateway Role	Control-plane enforcement node

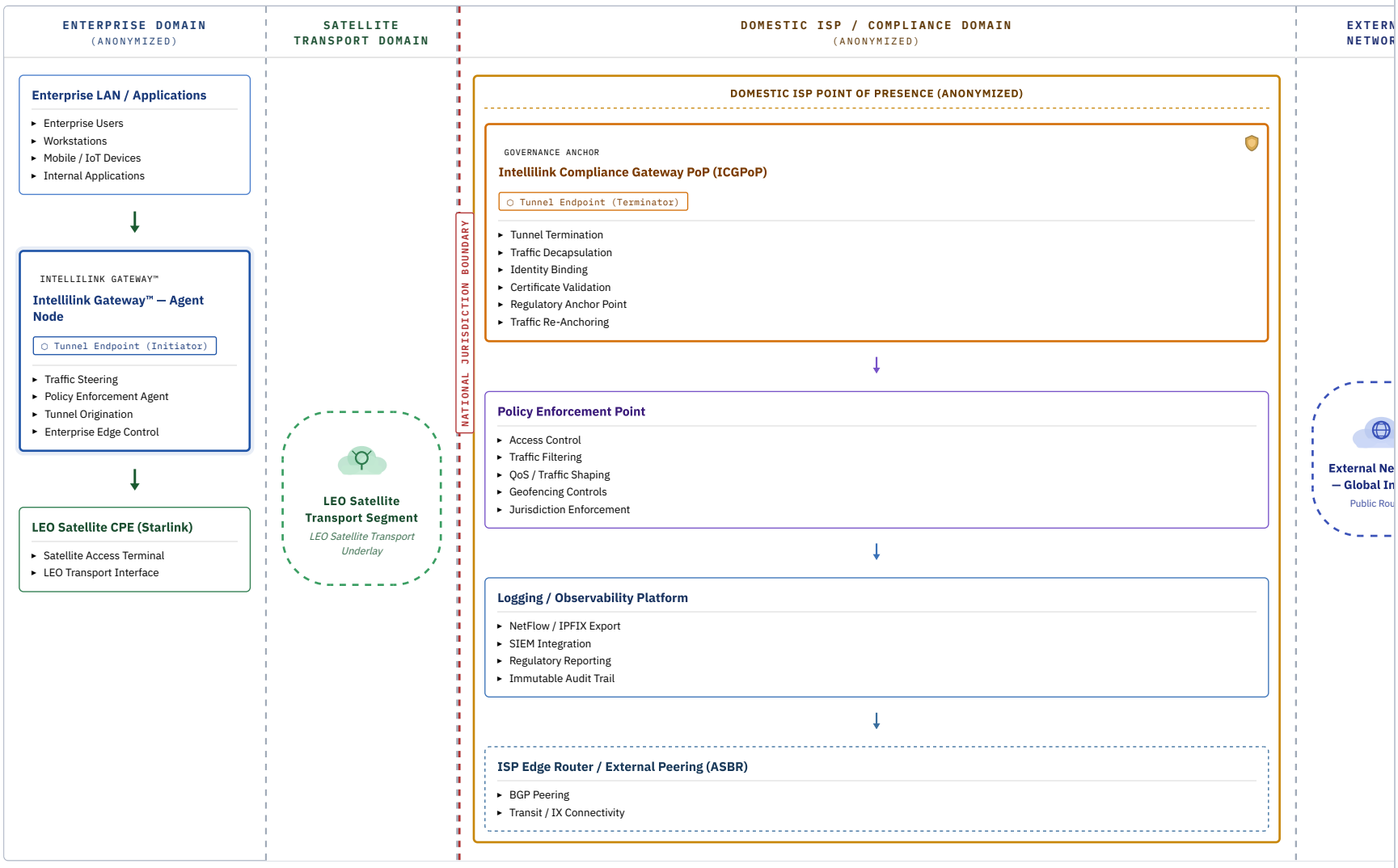
ANONYMIZATION NOTICE

All participating entities are anonymized. No enterprise customer data was retained beyond transient packet forwarding requirements.

TECHNICAL ARCHITECTURE REFERENCE · VALIDATION TEST WINDOW

Figure 3 — Anonymized Field Deployment Topology (Validation Test Window)

Enterprise satellite connectivity anchored through a domestic ISP compliance point to restore governance, visibility, and accountability.



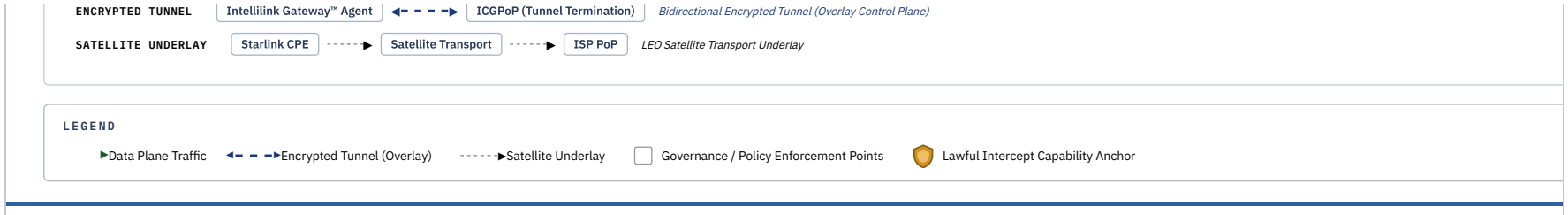


FIGURE 3 Anonymized Field Deployment Topology (Validation Test Window) — Enterprise traffic originates within the enterprise LAN and is steered by the Intellilink Gateway™ Agent through an encrypted overlay tunnel terminating at the Intellilink Compliance Gateway PoP (ICGPoP) within the domestic ISP domain. Upon tunnel termination, traffic undergoes decapsulation, identity binding, and policy enforcement before entering the ISP routing environment.

SECTION 05

Governance Capabilities Demonstrated

During validation, the following governance capabilities were demonstrated:

- **Accountable upstream presence:** enterprise traffic visibly enters an ISP PoP
- **Address sovereignty:** enterprise traffic appears under domestic IP addressing space
- **Traffic observability:** flow-level monitoring feasible at the PoP
- **Policy insertion point:** ISP-side controls possible without modification to satellite infrastructure
- **Audit readiness:** architecture supports integration with lawful intercept and logging systems where legally required

SECTION 06

What Was Proven

- ✔ Satellite connectivity can be governed without degrading performance
- ✔ ISP accountability can be restored without owning the satellite access link
- ✔ Enterprises can retain connectivity resilience while aligning with national frameworks
- ✔ The model can be deployed using existing ISP infrastructure

SECTION 07

What Was Explicitly Not Proven

- ✖ Commercial scalability
- ✖ Regulatory certification or approval
- ✖ Lawful intercept system integration (jurisdiction dependent)
- ✖ End-user billing or retail ISP functions

SCOPE NOTE

These areas are intentionally outside the scope of this validation exercise.

SECTION 08

Limitations & Next Validation Steps

Future evaluation may include:

- Multi-site enterprise deployments
- Formal lawful intercept integration (jurisdiction dependent)
- Long-term performance analytics
- Multi-satellite transport support

SECTION 08 — SUPPLEMENTARY REFERENCE

Control Plane vs Data Plane Separation

The following diagram illustrates the architectural separation between the data plane (traffic flow) and the control plane (overlay governance), clarifying where each layer operates, where policy is enforced, and where national jurisdiction re-applies.

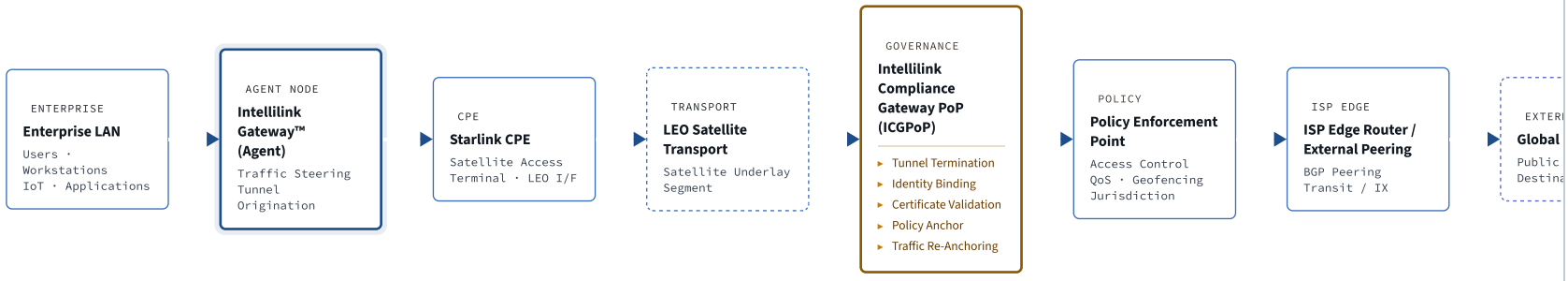
INTELLILINK GATEWAY™ · TECHNICAL ARCHITECTURE REFERENCE

Control Plane vs Data Plane Separation
Intellilink Gateway™ Architecture

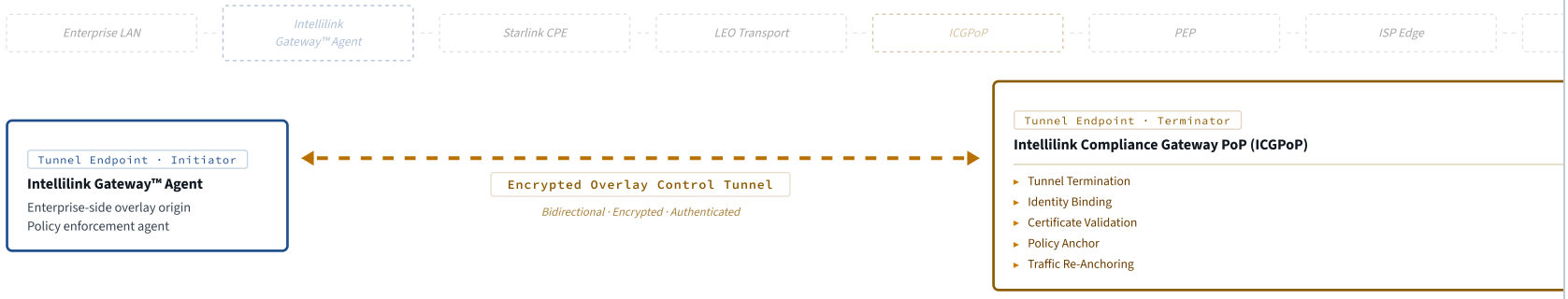
Illustrating traffic carriage, policy enforcement, and jurisdictional anchor points across the Intellilink overlay architecture.

DATA PLANE | CONTROL PLANE | REGULATOR REFERENCE

DATA PLANE — TRAFFIC FLOW



CONTROL PLANE — OVERLAY GOVERNANCE



The control plane tunnel traverses the satellite transport underlay transparently — it is not terminated or inspected at the CPE or satellite segment.

01 WHERE IS TRAFFIC CARRIED?

Enterprise LAN → Intellilink Agent → Starlink CPE → LEO Transport → ICGPoP → PEP → ISP Edge → Internet

Data Plane · Solid Arrow Path

02 WHERE IS POLICY APPLIED?

At the ICGPoP (identity binding, certificate validation, re-anchoring) and the Policy Enforcement Point (access control, QoS, jurisdiction).

ICGPoP + PEP · Governance Nodes

03 WHERE DOES JURISDICTION APPLY?

At the domestic ISP PoP. The ICGPoP serves as the nationa point — traffic is re-anchored before onward transit.

ICGPoP · National Jurisdiction Boundary

FIGURE 4 Control Plane vs Data Plane Separation — Intellilink Gateway™ Architecture. Illustrating traffic carriage, policy enforcement, and jurisdictional anchor points across the overlay architecture.

SECTIONS 09 – 10

LEGAL & OPERATIONAL DISCLAIMER

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The Intellilink Gateway™ control plane operates independently of satellite providers and does not imply endorsement or partnership with any satellite operator.

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